

Documentation

For VLANs on 2 IOU Switches

```

IOU7#show running-config
Building configuration...

Current configuration : 3177 bytes
!
  Last configuration change at 19:06:51 UTC Wed Nov 26 2025
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname IOU7
!
boot-start-marker
boot-end-marker
!
!
logging discriminator EXCESS severity drops 6 msg-body drops EXCESSCOLL
logging buffered 50000
logging console discriminator EXCESS
!
no aaa new-model
!
!
no ip icmp rate-limit unreachable
!
!
no ip domain-lookup
no ip cef
no ipv6 cef
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
ip tcp synwait-time 5
!

```

This section contains the basic global settings of the IOU7 switch. It includes system version, timestamps for logging, hostname, logging configuration, Spanning-Tree Rapid-PVST setup, and internal VLAN allocation policy. These initial settings prepare the switch for VLAN and interface configurations later in the project.

```

interface Ethernet1/0
  switchport access vlan 30
  switchport mode access
  spanning-tree portfast edge
!
interface Ethernet1/1
  switchport access vlan 30
  switchport mode access
  spanning-tree portfast edge
!
interface Ethernet1/2
  switchport access vlan 40
  switchport mode access
  spanning-tree portfast edge
!
interface Ethernet1/3
  switchport access vlan 40
  switchport mode access
  spanning-tree portfast edge
!
interface Ethernet2/0
  switchport access vlan 50
  switchport mode access
  spanning-tree portfast edge
!
interface Ethernet2/1
  switchport access vlan 50
  switchport mode access
  spanning-tree portfast edge
!
interface Ethernet2/2
  switchport access vlan 60
  switchport mode access
  spanning-tree portfast edge
!
interface Ethernet2/3
  switchport access vlan 60
  switchport mode access
  spanning-tree portfast edge
!
interface Ethernet3/0
  description Trunk-to-Fortigate
  switchport trunk allowed vlan 10,20,30,40,50,60
  switchport trunk encapsulation dot1q
  switchport mode trunk
  spanning-tree portfast edge trunk
!
interface Ethernet3/1
  switchport access vlan 60
  switchport trunk encapsulation dot1q
  switchport mode access
  spanning-tree portfast edge trunk
!

```

This configuration assigns access ports to VLANs 30, 40, 50 and 60 (for end devices) with Spanning-Tree PortFast enabled to speed host connectivity, and configures Ethernet3/0 as a dot1q trunk to the FortiGate carrying VLANs 10–60. Note: Ethernet3/1 contains conflicting trunk/access commands and should be removed.

```

192.168.110.132 - PuTTY
description Trunk-to-Fortigate
switchport trunk allowed vlan 10,20,30,40,50,60
switchport trunk encapsulation dot1q
switchport mode trunk
spanning-tree portfast edge trunk
!
interface Ethernet3/1
switchport access vlan 60
switchport trunk encapsulation dot1q
switchport mode access
spanning-tree portfast edge trunk
!
interface Ethernet3/2
!
interface Ethernet3/3
!
interface Vlan1
no ip address
shutdown
!
interface Vlan10
ip address 192.168.10.1 255.255.255.0
ip helper-address 192.168.60.10
!
interface Vlan20
ip address 192.168.20.1 255.255.255.0
ip helper-address 192.168.60.10
!
interface Vlan30
ip address 192.168.30.1 255.255.255.0
ip helper-address 192.168.60.10
!
interface Vlan40
ip address 192.168.40.1 255.255.255.0
ip helper-address 192.168.60.10
!
interface Vlan50
ip address 192.168.50.1 255.255.255.0
ip helper-address 192.168.60.10
!
interface Vlan60
ip address 192.168.60.1 255.255.255.0
ip helper-address 192.168.60.10
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!

```

This section continues the configuration of the trunk ports and sets up VLAN interfaces (SVIs) with IP addresses for routing between VLANs. VLANs 10, 20, 30, 40, 50, and 60 are assigned IPs and a DHCP helper address pointing to 192.168.60.10 for centralized DHCP services. VLAN1 is shut down since it is not used. Basic settings like IP forwarding and disabling HTTP servers are applied to secure the switch. Ethernet3/0 remains the main trunk to FortiGate, while Ethernet3/1 still has conflicting trunk/access commands that should be corrected.

```

!
!
!
IOU2#show running-config
Building configuration...

Current configuration : 2338 bytes
!
Last configuration change at 19:06:51 UTC Wed Nov 26 2025
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname IOU2
!
boot-start-marker
boot-end-marker
!
logging discriminator EXCESS severity drops 6 msg-body drops EXCESSCOLL
logging buffered 50000
logging console discriminator EXCESS
!
no aaa new-model
!
!
no ip icmp rate-limit unreachable
!
!
no ip domain-lookup
no ip cef
no ipv6 cef
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
ip tcp synwait-time 5
!
!

```

This section shows the basic global settings of the switch IOU2. It includes system version, timestamping for logs, hostname, logging configuration, Spanning-Tree Rapid-PVST setup, and internal VLAN allocation policy. These settings prepare the switch for VLAN and interface configurations later in the project, similar to the IOU7 setup.

```

interface Ethernet0/0
description Trunk-to-Fortigate
switchport trunk allowed vlan 10,20,30,40,50
switchport trunk encapsulation dot1q
switchport mode trunk
spanning-tree portfast edge trunk
!
interface Ethernet0/1
switchport access vlan 10
switchport mode access
spanning-tree portfast edge
!
interface Ethernet0/2
switchport access vlan 10
switchport mode access
spanning-tree portfast edge
!
interface Ethernet0/3
switchport access vlan 20
switchport mode access
spanning-tree portfast edge
!
interface Ethernet1/0
switchport access vlan 20
switchport mode access
spanning-tree portfast edge
!
interface Ethernet1/1
switchport access vlan 30
switchport mode access
spanning-tree portfast edge
!
interface Ethernet1/2
switchport access vlan 30
switchport mode access
spanning-tree portfast edge
!
interface Ethernet1/3
switchport access vlan 40
switchport mode access
spanning-tree portfast edge
!
interface Ethernet2/0
switchport access vlan 40
switchport mode access
spanning-tree portfast edge
!
interface Ethernet2/1
switchport access vlan 50
switchport mode access
spanning-tree portfast edge
!

```

This configuration assigns the switch ports to their respective VLANs for end devices and sets up a trunk link to the FortiGate firewall. Ethernet0/0 is configured as a dot1q trunk carrying VLANs 10, 20, 30, 40, and 50. The rest of the ports are configured as access ports with Spanning-Tree PortFast enabled, distributing devices across VLANs 10, 20, 30, 40, and 50 according to the network design.

```

192.168.110.132 - PuTTY
interface Ethernet2/1
switchport access vlan 50
switchport mode access
spanning-tree portfast edge
!
interface Ethernet2/2
switchport access vlan 50
switchport mode access
spanning-tree portfast edge
!
interface Ethernet2/3
!
interface Ethernet3/0
!
interface Ethernet3/1
!
interface Ethernet3/2
!
interface Ethernet3/3
!
interface Vlan1
no ip address
shutdown
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
!
!
!
control-plane
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
!
!
end
IQU2#

```

This section completes the port assignments for VLAN 50 and leaves the remaining ports unconfigured. VLAN1 is shut down as it is not used. Basic switch settings include enabling IP forwarding, disabling HTTP servers for security, and configuring console, auxiliary, and VTY lines with full privileges and logging synchronization. These settings finalize the switch preparation for network operations.